

The charge q on the capacitor at any instant is

$$q = C v_c$$

output voltage $v_{out} = iR = \frac{dq}{dt} R = CR \frac{dv_c}{dt} = CR \frac{dv_i}{dt}$

Semiconductor:

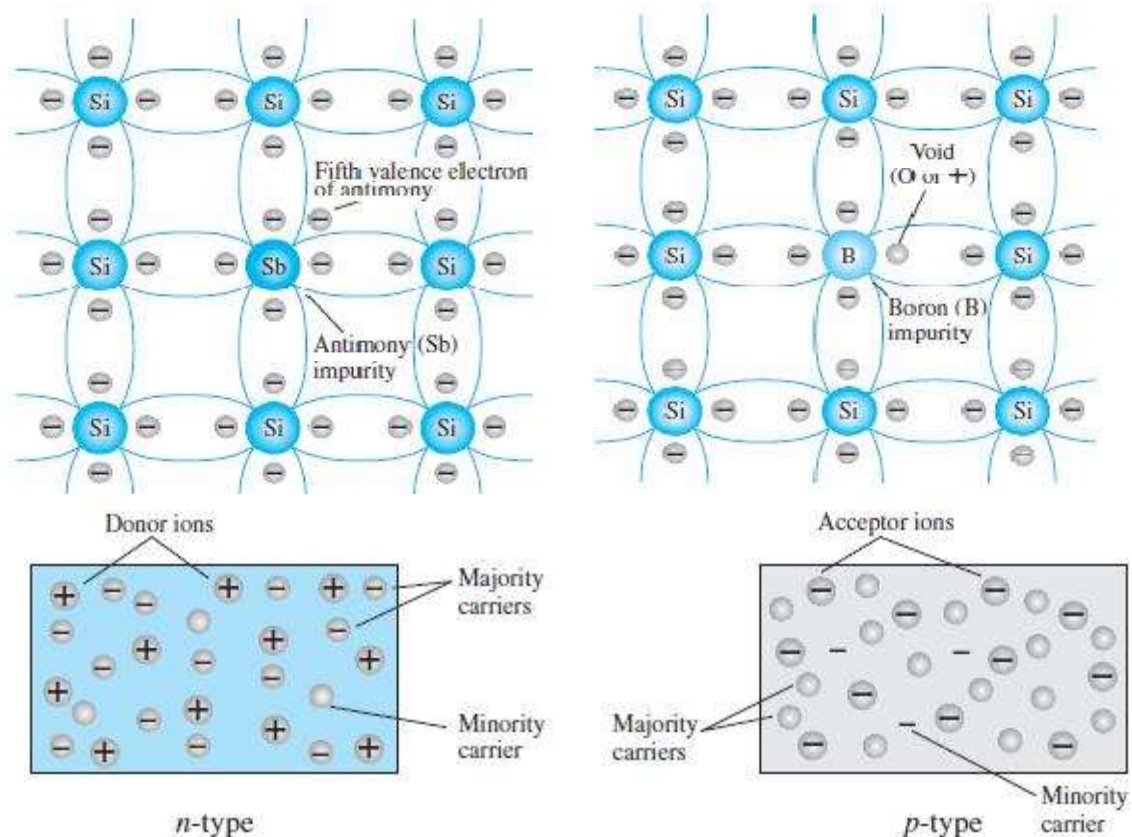
- Conductivity lies between conductor and insulator.
- Forbidden energy gap 0.2-2.5eV.
- At 0K a pure semiconductor behaves as an insulator.
- Semiconductor materials show a reduction in resistance with increase in temperature. So said to have a negative temperature coefficient.

Intrinsic Semiconductor:

- Semiconductor refined to reduce the number of impurities to a very low level.
e.g : Semiconductor in pure form
- Group-IV elements. Si, Ge,

Extrinsic Semiconductor:

- To increase the conductivity, impurities also called dopant (Group III or V) are added to the pure semiconductor material and is called extrinsic Semiconductor (n-type or p-type). The process is called doping.
- N-type Semiconductor- Pentavalent (As, Sb, P) atom is added to pure semiconductor. Diffused impurities with five valence electrons are called donor atoms.
- P-type Semiconductor- Trivalent (Al, B, Ga) atom is added to pure semiconductor. Diffused impurities with three valence electrons are called acceptor atoms.

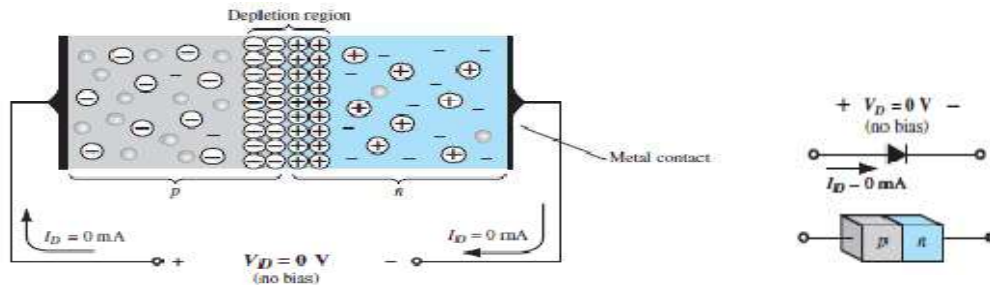


- Holes are the majority carrier in p-type semiconductor and electrons are minority carrier. In n-type semiconductor electrons are the majority carrier and holes are the minority carrier.

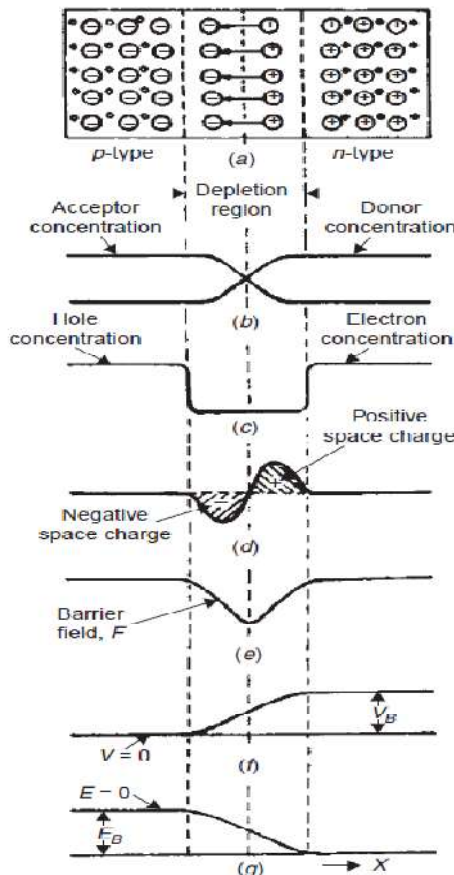
Diode:

- Solid state device created by joining the p-type and n-type material is called as semiconductor diode.

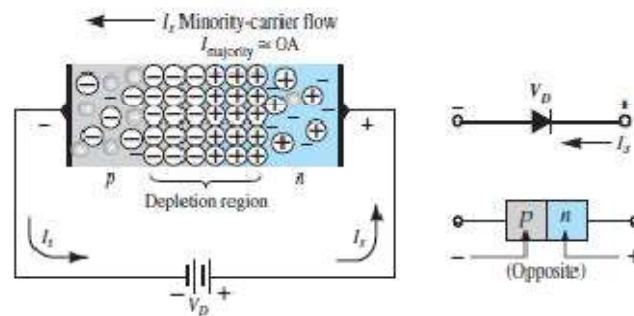
No Bias ($V=0$)



- Absence of external voltage across the p-n junction is called the unbiased diode. Because of the density gradient electrons and holes diffuse and they combine leaving the ions unneutralised and are called uncovered charges.
- The uncovered charges generate an electric field directed from n-side to p-side called as barrier field which opposes the diffusion process further.
- Since the vicinity of the junction is depleted of mobile charges. Hence called as depletion region.

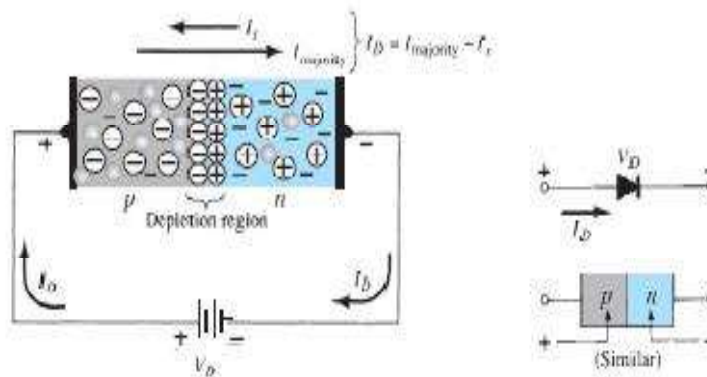


Reverse Bias ($V_D < 0V$)



- Positive polarity of the external bias V_D is connected to n-type and negative terminal is connected to p-type.
- The number of uncovered positive and negative ions will increase in the depletion region causing widening the depletion region which creates a great barrier for the majority carrier to overcome, effectively reducing the majority carrier flow to zero and hence the current due to majority carrier $I_{\text{majority}}=0$
- The minority carriers which travels down the potential barrier remain unaffected and give a small current called the reverse saturation current denoted as I_s .

Forward Bias ($V_D > 0V$)



- Positive polarity of the external bias V_D is connected to p-type and negative terminal is connected to n-type.
- External bias V_D exerts a force on the mobile carriers to move them towards the junction. At the boundary they recombine with the ions and reduce the width of the depletion region.
- The depletion region will continue to decrease in width as the voltage is increased further and a heavy flood of electrons will move from n-side to p-side giving the I_{majority} an exponential rise from p-side to n-side,
- The minority carrier flow will not be affected by this because the conduction level is determined by the limited number of impurities in the material and the current is

denoted by I_s .

The total current is given by

$$I_D = I_{\text{Forward}} + I_{\text{Reverse}} \\ = I_{\text{majority}} - I_{\text{minority}} \text{ (Direction opposite)}$$

In terms of reverse saturation current, I_D can be written as

$$I_D = I_s \exp\left(\frac{eV}{\eta KT}\right) - I_s \text{ is called the Shockley's equation.}$$

Where

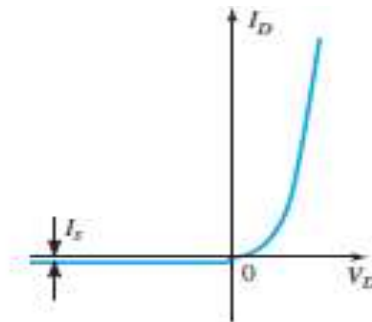
e- Charge of an electron

K- Boltzmann's Constant

T- Temperature in Kelvin

η - Quality factor depends upon the diode material ($\eta=2$ for Si and 1 for Ge)

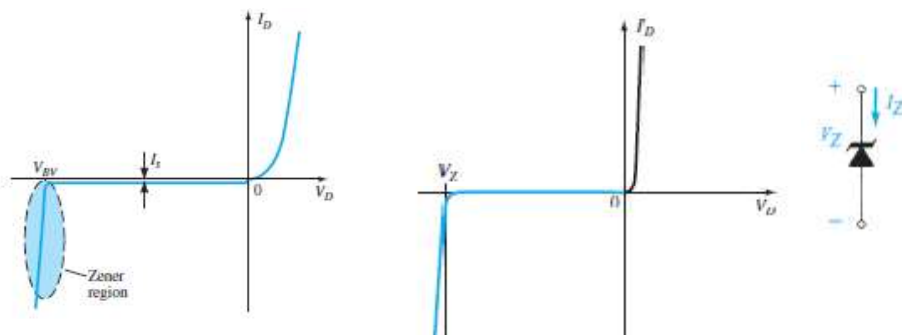
V- Supplied voltage across the junction



Breakdown Condition:

(a) Zener Breakdown

- Too much of reverse bias across a p-n junction exerts a strong force on a bound electron to tear it out from the covalent bond. Thus a large number of electron and hole pairs will be generated through a direct rupture of the covalent bonds and they increase the reverse current and give a sharp increase in the characteristics. It is called Zener breakdown. Diode employing the unique portion of the characteristics of a p-n junction is called Zener diode.
- Maximum reverse voltage potential that can be applied before entering the Zener region is called the peak inverse voltage (PIV) or peak reverse voltage (PRV).

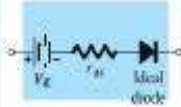
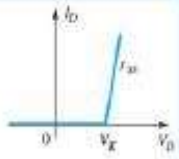
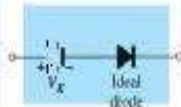
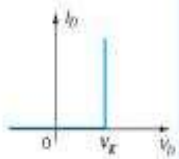
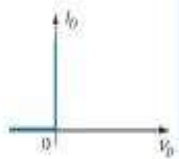


(b) Avalanche Breakdown:

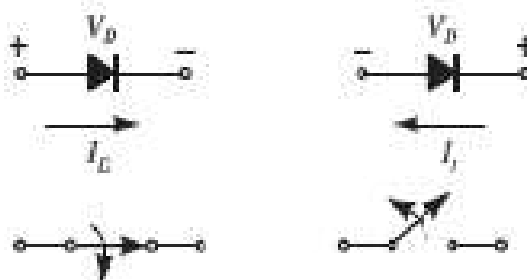
- With increasing reverse bias voltage, the electric field across the junction of a diode increases. At a certain value of the reverse voltage, the electric field imparts a sufficiently high energy to a thermally generated carrier. The carrier on colliding with an ion on its way disrupts a covalent bond and gives a new hole electron pair. This process is cumulative and gives an avalanche of carriers in a very short time. It is called avalanche multiplication.

Diode equivalent Circuit:

- Equivalent circuit is a combination of element properly chosen to best represent the actual terminal characteristics of a device or system in a particular operating point.

Diode Equivalent Circuits (Models)			
Type	Conditions	Model	Characteristics
Piecewise-linear model			
Simplified model	$R_{\text{network}} \gg r_{av}$		
Ideal device	$R_{\text{network}} \gg r_{av}$ $E_{\text{network}} \gg V_K$		

Ideal diode in forward and reverse biased condition is as follows



Diode Resistance levels:

- According to the applied signal the resistance levels in a diode has following type
 1. DC or Static (DC signal)
 2. AC or Dynamic (Small AC signal)
 3. Average ac (Large AC signal)

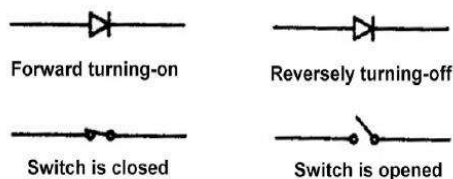
Type	Equation	Special Characteristics	Graphical Determination
DC or static	$R_D = \frac{V_D}{I_D}$	Defined as a point on the characteristics	
AC or dynamic	$r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{26 \text{ mV}}{I_D}$	Defined by a tangent line at the Q-point	
Average ac	$r_{av} = \frac{\Delta V_d}{\Delta I_d} \Big _{\text{pt. to pt.}}$	Defined by a straight line between limits of operation	

CLIPPER

It controls the shape of the output waveform by removing or clipping a portion of the applied wave. Half wave rectifier is the simplest example. It is also referred as voltage limiters/ amplitude selectors/ slicers.

Applications:

- In radio receivers for communication circuits.
- In radars, digital computers and other electronic systems.
- Generation for different waveforms such as trapezoidal, or squarewaves. Helps in processing the picture signals in television transmitters.
- In television receivers for separating the synchronizing signals from composite picture signals



Types of Clipper Circuit

1. Series- Diode is in series with the source
 2. Parallel- Diode is in parallel with the source.
- Clipper circuit which uses a DC battery is called a biased clipper.

SERIES CLIPPER:

Assumption- diode is ideal in characteristics

Analysis

+ve Half Cycle:

Diode is on because of forward biasing condition. Since no voltage drop across the diode the output voltage becomes

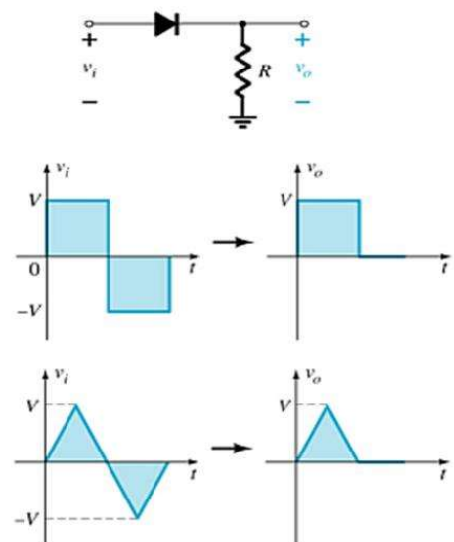
$$V_O = V_R = V_i$$

-ve Half Cycle:

Diode is off because of reverse biasing condition.

Since no current flows through the circuit the output voltage $V_O = 0$.

Figure shows the output waveform of a simple series clipper with input as square and triangular waveform. Since the negative half cycle is clipped off in the output it is called as a negative clipper circuit.



Biased Series Clipper:

Assumption- diode is ideal in characteristics

Analysis

Since the diode is on because of the 5v battery
The transition of the diode from one state to another can be found out to be at $V_i = -5\text{V}$ above which the diode is ON and below which the diode is OFF.

+ve Half Cycle:

Since the diode is on the output voltage will be (Applying KVL)

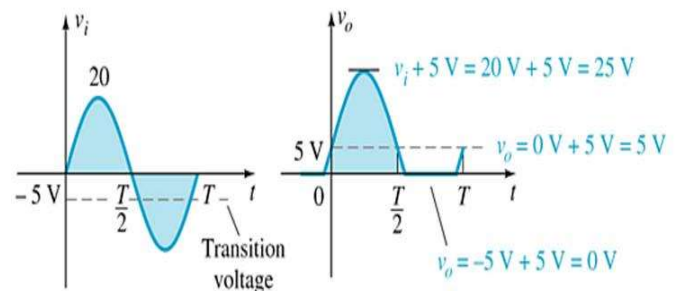
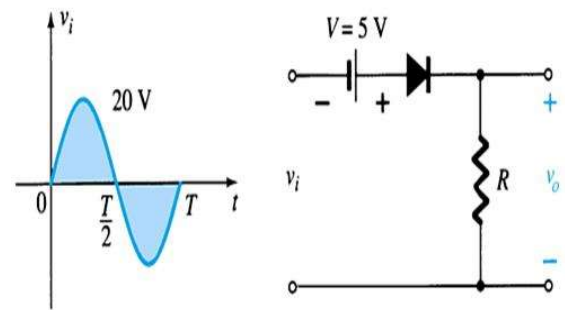
$$V_i + 5 = V_R$$

$$V_O = V_i + 5$$

-ve Half Cycle:

Since the diode is off $V_O = 0$.

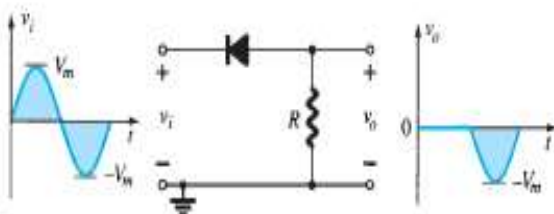
Figure Shows the input and output waveform.



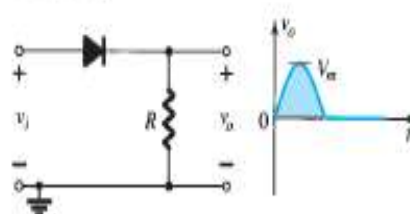
Example of Other Series Clipper Circuits:

Simple Series Clippers (Ideal Diodes)

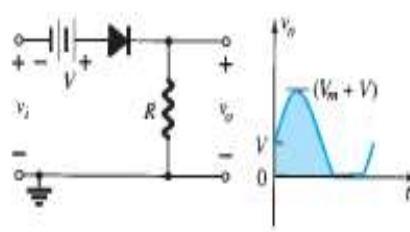
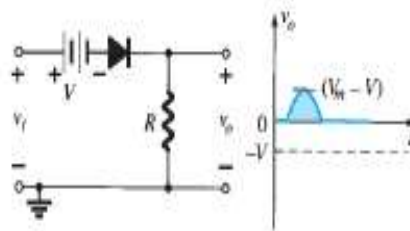
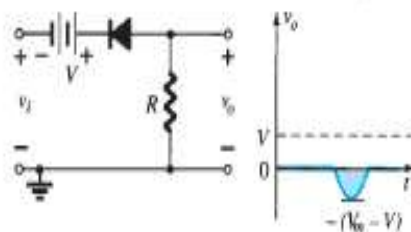
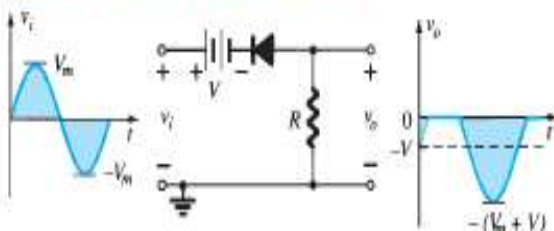
POSITIVE



NEGATIVE



Biased Series Clippers (Ideal Diodes)



PARALLEL CLIPPER:

Assumption- diode is ideal in characteristics

Analysis

+ve Half Cycle:

Diode is on because of forward biasing condition.
Since no voltage drop across the diode the output voltage becomes

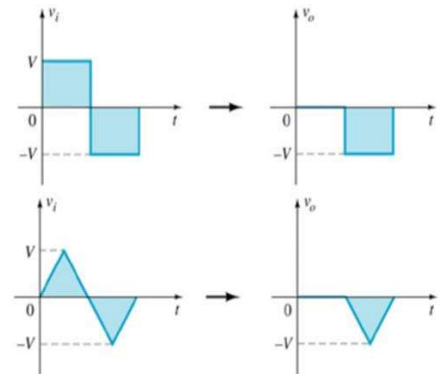
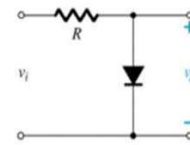
$$V_O = V_d = 0$$

-ve Half Cycle:

Diode is off because of reverse biasing condition.

Since no current flows through the circuit the output voltage $V_O = V_i$.

Figure shows the output waveform of a simple parallel clipper with input as square and triangular waveform. Since the positive half cycle is clipped off in the output it is called as a positive clipper circuit.



Biased parallel Clipper:

Assumption- diode is ideal in characteristics

Analysis

The transition of the diode from one state to another can be found out to be at $V_i = 4\text{V}$ above which the diode is OFF and below which the diode is ON.

+ve Half Cycle:

Since the diode is OFF (above 4V) the output voltage will be (Applying KVL)

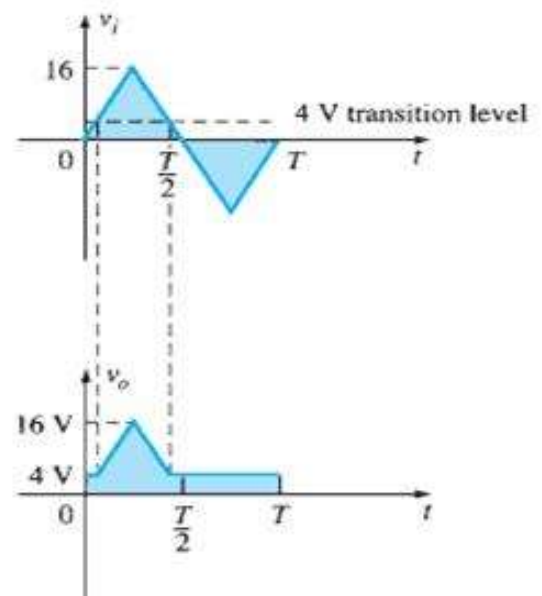
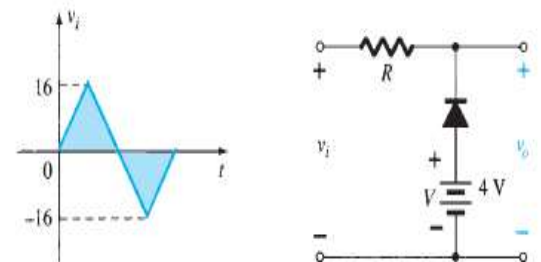
$$V_i = V_o$$

-ve Half Cycle:

Since the diode is ON (below 4V)

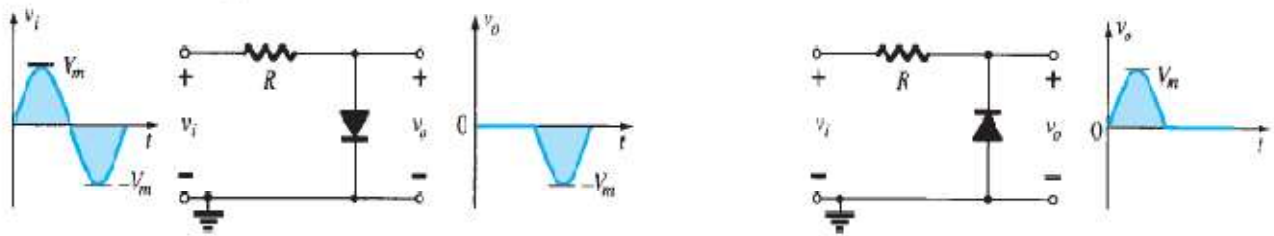
$$V_O = 4\text{V}.$$

Figure Shows the input and output waveform.

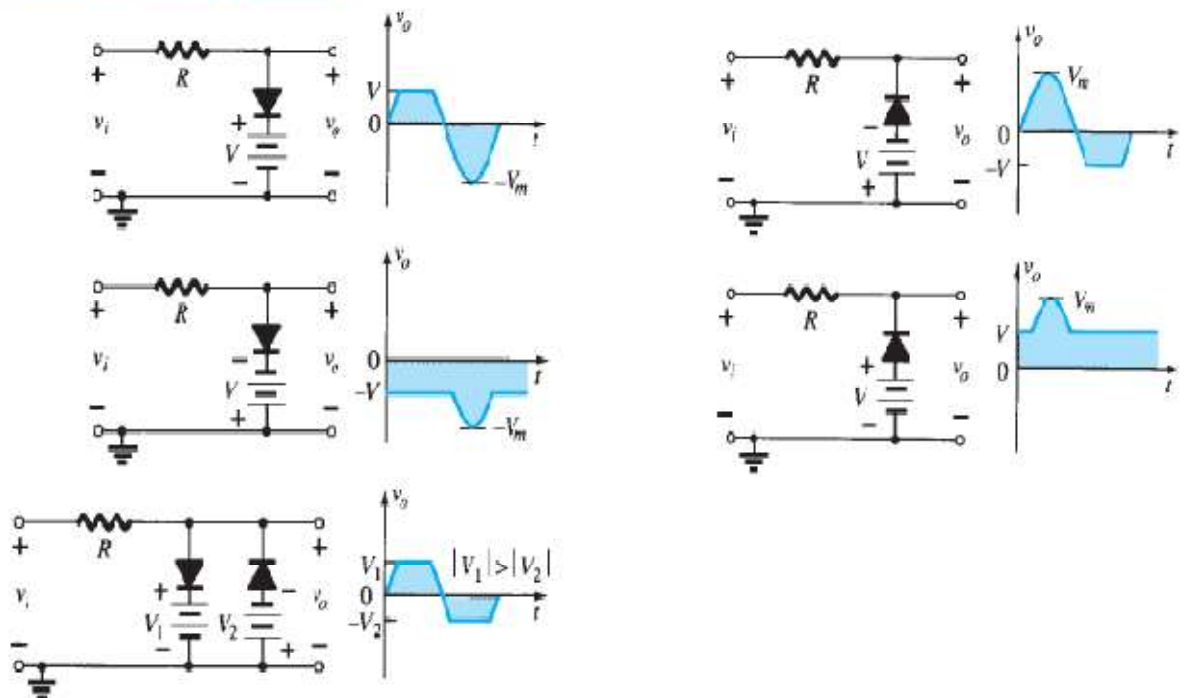


Example of Other Parallel Clipper Circuits:

Simple Parallel Clippers (Ideal Diodes)



Biased Parallel Clippers (Ideal Diodes)



CLAMPERS:

- A diode and capacitor can be combined to “clamp” an AC signal to a specific DC level.
- It must have a capacitor, a diode and a resistive element.
- For additional shift an independent DC supply can be introduced in the circuit.
- The time constant $\tau=RC$ must be large enough to ensure that the voltage across the capacitor does not discharge significantly during the diode is nonconducting.

Procedure to analyze a clamper circuit

1. Consider the part of the input signal that will forward bias the diode.

2. During the On state assume that the capacitor will charge up instantaneously to a voltage level determined by the network.
3. Assume that during the diode is in OFF state the capacitor will hold on to its established voltage level.
4. The polarity of V_o must be same throughout the analysis.
5. Total swing of the total output must match the swing of the input signal.

-ve clamper analysis:

-Diode is ON(Short Circuit) in the positive half cycle.

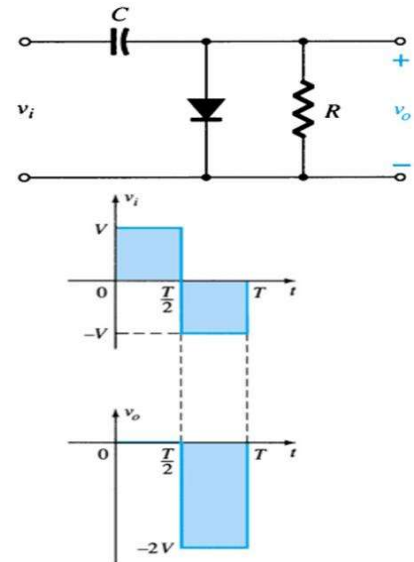
- Established voltage level on the capacitor

$$V_c = V$$

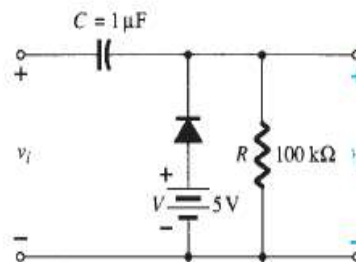
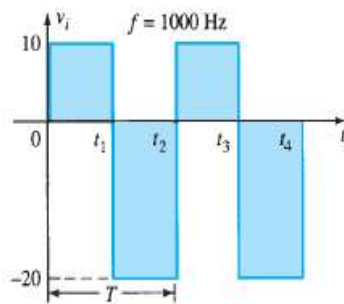
-During negative half cycle the diode is OFF and the output voltage is

$$V_o = V_R = -V_i - V_c = -2V$$

-Total swing of output is $-2V$ which is same as the total swing of the input.



Biased Clamper Circuit:



-ve half cycle:

Diode is ON (S.C). So applying KVL around the input loop we have

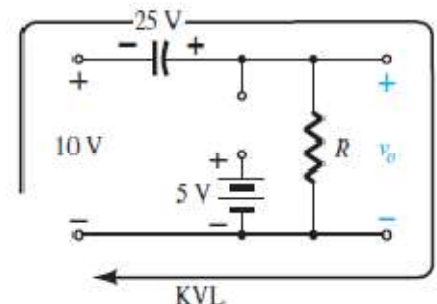
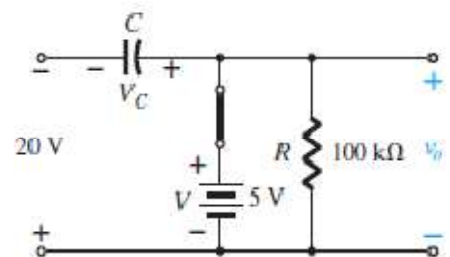
$$-20 + V_c = 5 = 0$$

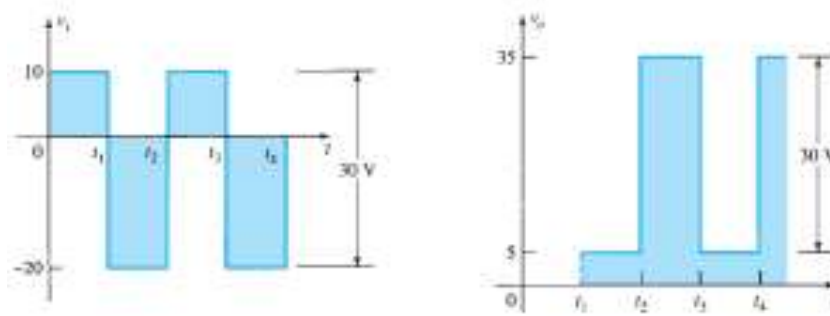
$$V_c = 25V \text{ and } V_o = 5V$$

During +ve half cycle diode is OFF. Applying KVL around the outside loop we have

$$10 + 25 - V_o = 0$$

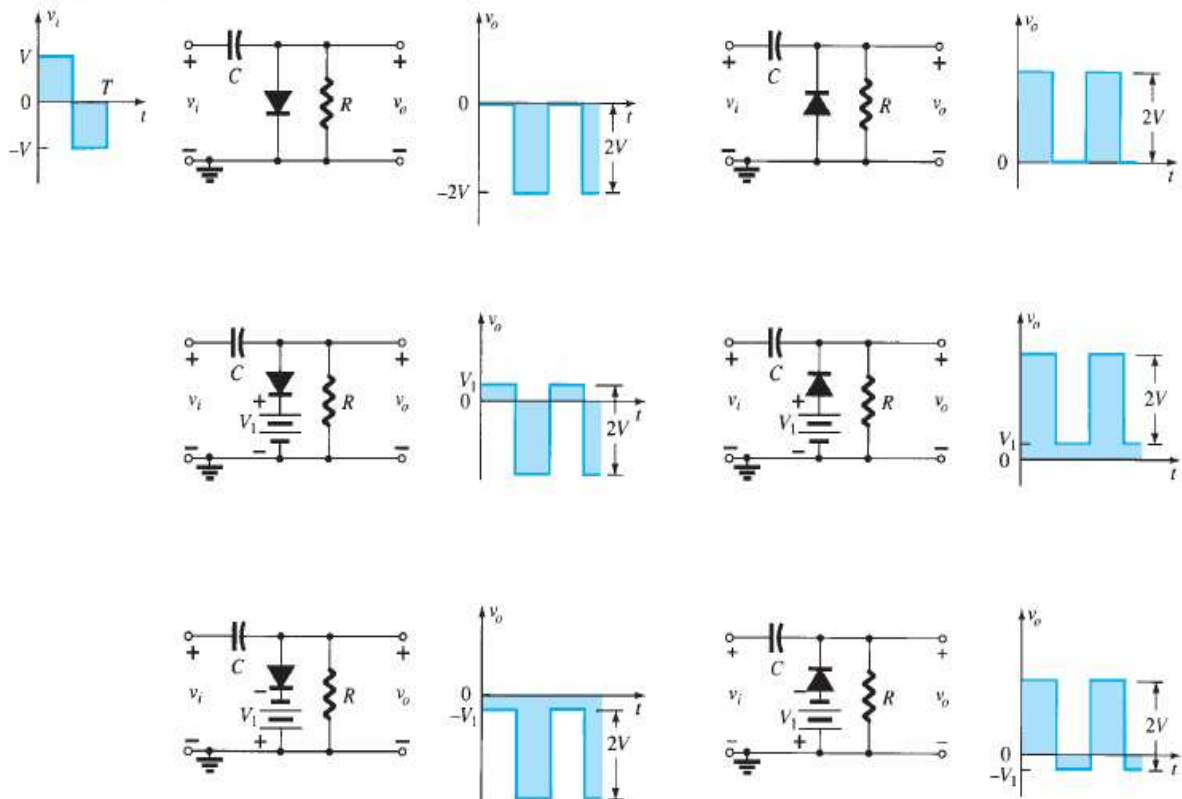
$$V_o = 35V$$





Summary of the Clamper Circuit:

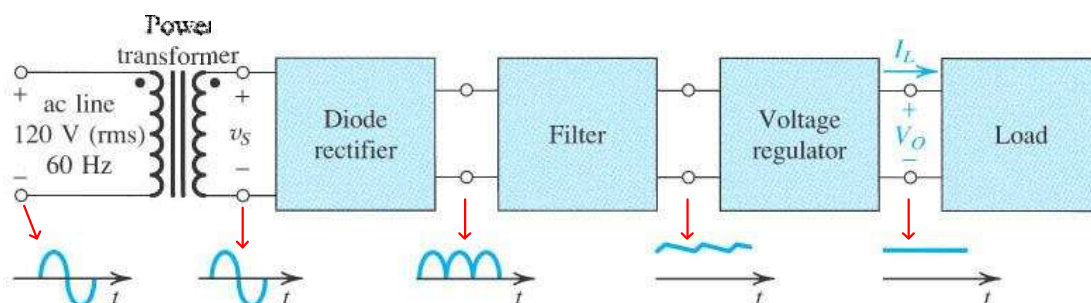
Clamping Networks



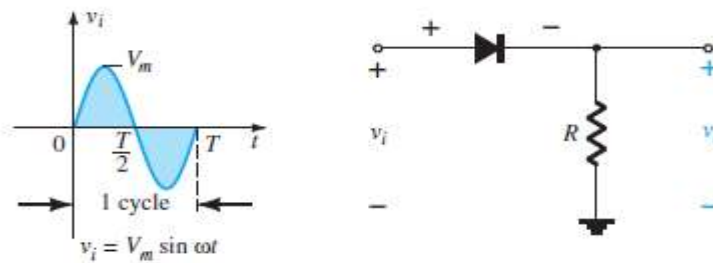
RECTIFIERS

An important application of “regular” diodes is in rectification circuits. These circuits are used to convert AC signals to DC in power supplies.

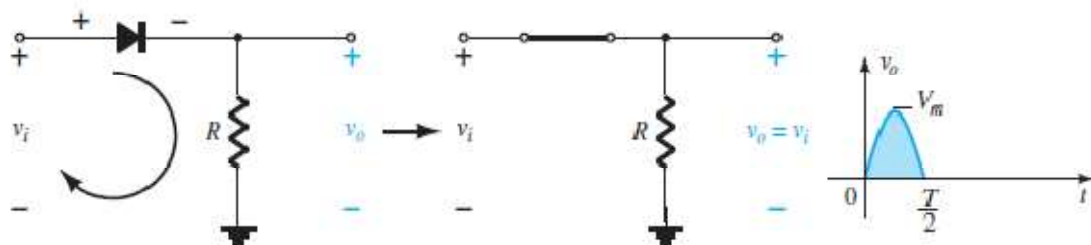
A block diagram of this process in a DC power supply is shown below.



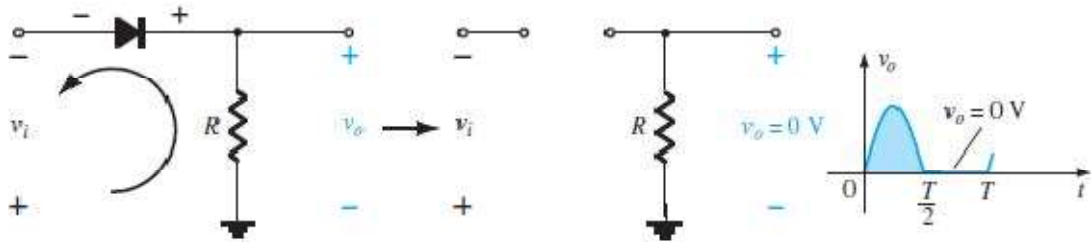
Half-Wave Rectifier:



The above circuit is called as a Half-wave rectifier since it will generate a waveform v_o that will have an average value of particular use in the ac-to-dc conversion process.



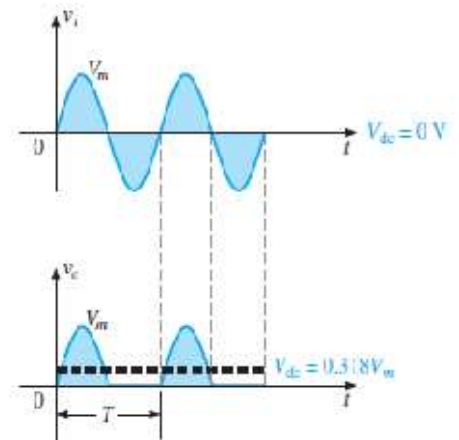
During $0 \rightarrow T/2$ (Positive Half Cycle) the diode is ON. Assuming an ideal diode with no voltage drop across it the output voltage v_o will be $v_o = V_R = V_m$



During $T/2 \rightarrow T$ (Negative Half Cycle) the diode is OFF (Open Circuit). So the current flowing through the circuit will be 0. The output voltage v_o will be

$$v_o = V_R = i \times R = 0$$

Figure shows the input and output waveform with output $V_{dc} = 0.318 V_m$.



Disadvantage:

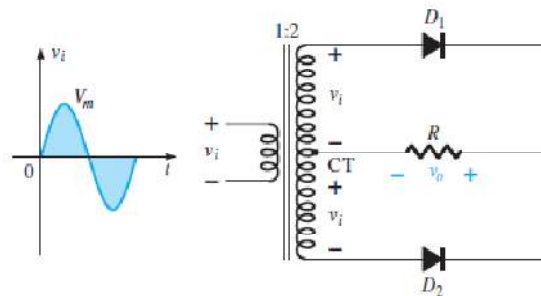
1. The ac supply delivers power only half the time.
2. Pulsating current frequency is equal to the supply frequency.

Full wave Rectifier:

The full wave rectifier utilizes both the positive and negative portions of the input waveform. Types of full wave rectifier are

- (a) Centre tapped configuration
- (b) Bridge configuration

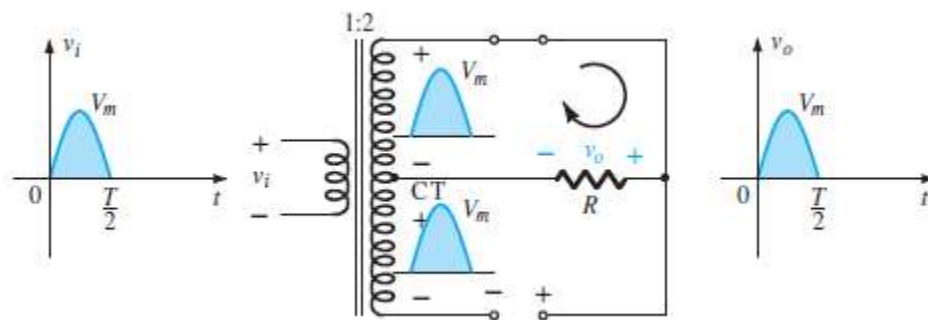
Centre tapped configuration:



- Current flows through the load resistance in the same direction during the full cycle of the input signal.
- Centre tap transformer is used with the secondary winding.

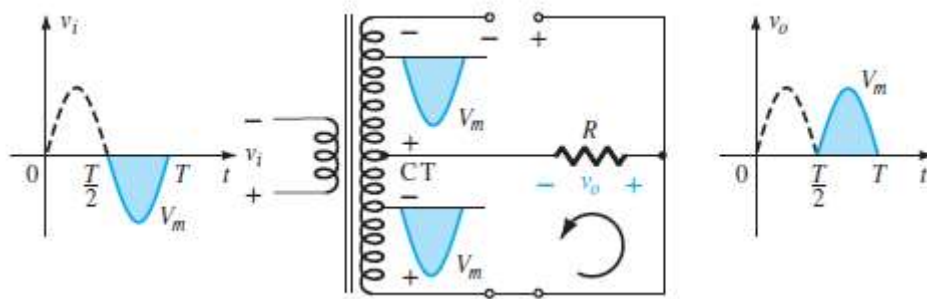
+ve Half Cycle:

- Diode D_1 is short circuited and D_2 is open circuited. Current flows through the upper half of the secondary winding.

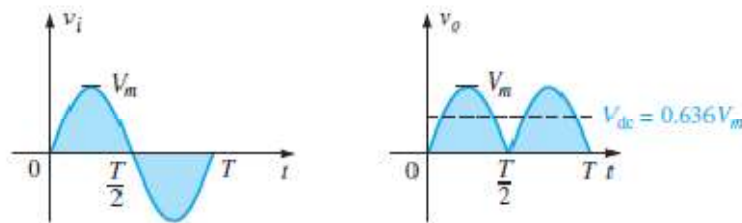


-ve Half Cycle:

- Diode D_2 is short circuited and D_1 is open circuited. Current flows through the lower half of the secondary winding.



Complete input and output waveform can be shown as



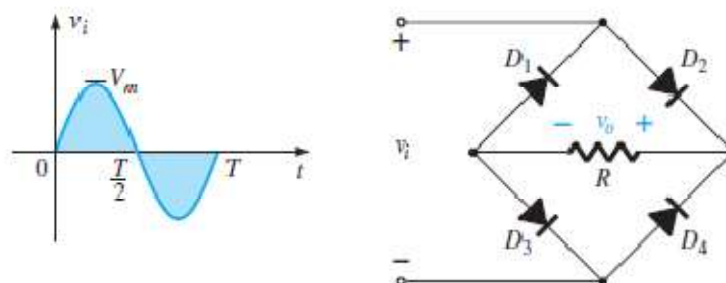
While this full-cycle rectifier is a big improvement over the half-cycle, there are some disadvantages.

Disadvantages:

- It is difficult to locate the centre tap on the secondary winding.
- The diodes must have high PIV.

Bridge Rectifier:

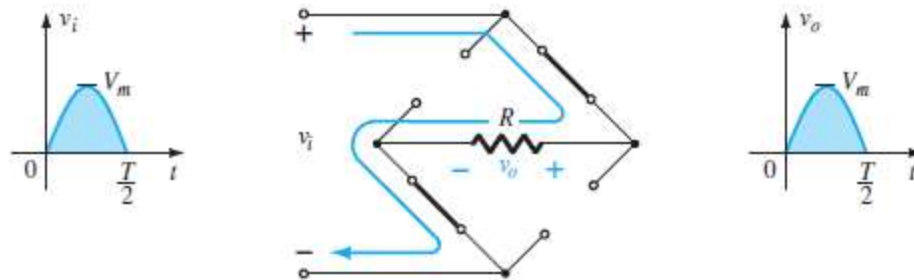
The bridge rectifier uses four diodes connected in bridge pattern.



The operation of the bridge rectifier can be summarized as:

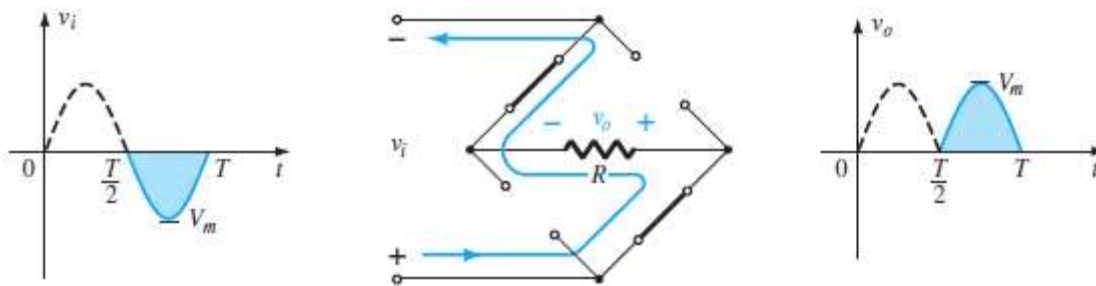
+ve Half Cycle:

- Diode D_1 and D_3 are short circuited and D_2 and D_4 are open circuited. Current flows through D_1 and D_3 to give the output voltage across the resistor.

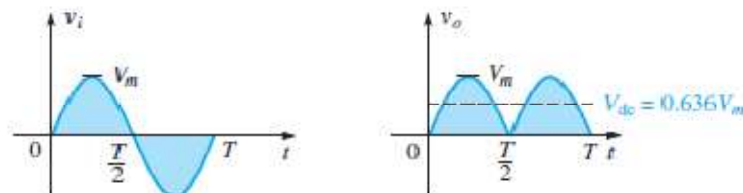


-ve Half Cycle:

- Diode D_1 and D_3 are open circuited and D_2 and D_4 are short circuited. Current flows through D_2 and D_4 to give the output voltage across the resistor.



Complete input and output waveform can be shown as



Advantages:

- No centre tapped transformer is required.
- PIV is less.

Disadvantages:

- It requires four diode and the power loss in the rectifier element is more.

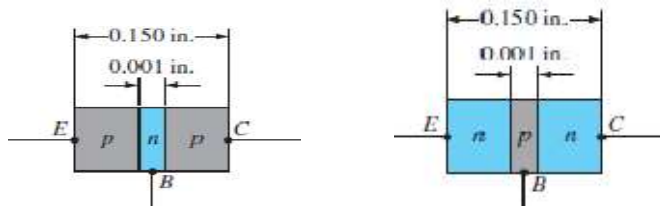
MODULE-II

TRANSISTOR:

1. Bipolar Junction Transistor (BJT)
2. Field Effect Transistor (FET)

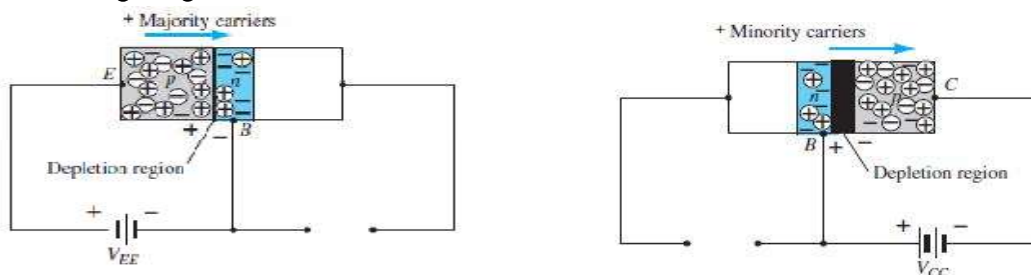
Bipolar Junction Transistor (BJT):

- pnp-- n-type semiconductor is sandwiched between two p-type semiconductor.
- npn-- p-type semiconductor is sandwiched between two n-type semiconductor.
- It has three terminal naming E-Emitter, B-Base and C-Collector.
- Both electron and hole responsible for the current conduction. So called bipolar junction transistor.
- Doping wise-Emitter>Collector>Base
- Two Junctions- J_{EB} (Junction emitter base) and J_{CB} (Junction collector base).



Transistor Operation:

- J_{EB} is forward biased by the battery V_{EE} by which the depletion region will decrease and a majority carrier flow will occur from emitter to base giving current I_{majority} or I_E .
- J_{CB} is reverse biased by the battery V_{CC} by which the depletion region will increase and a minority carrier flow will occur from base to collector giving current I_{minority} .
- When both the battery supplies are given simultaneously the holes in the base region due to the battery V_{EE} will act as minority carrier. They will cross the base region to reach the collector giving the current I_C .



So the current equations of BJT can be written as

$$I_E = I_C + I_B$$

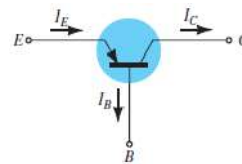
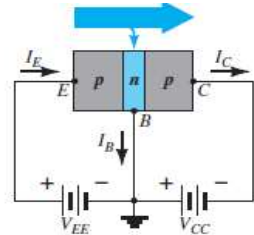
$$I_C = I_{\text{majority}} + I_{\text{minority}}$$

$$I_C = \alpha I_E + I_{C0}$$

Where α is defined as the fraction of the total emitter current that represents holes which have travelled from emitter across the base to the collector and I_{C0} is called as leakage current.

- Depending upon the common terminal between input and output circuit of a transistor it may be operated in 3 modes of a BJT
 - Common Base
 - Common Emitter
 - Common Collector

Common Base:



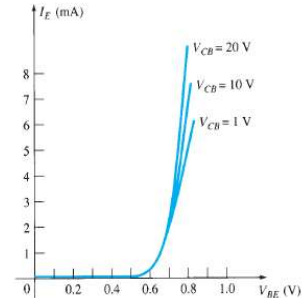
- Depending upon the biasing of the J_{EB} and J_{CB} , transistor has three region of operation.

	J_{EB}	J_{CB}	Region of operation	Application
	Forward Bias	Forward Bias	Saturation	ON Switch
	Forward Bias	Reverse Bias	Active	Amplifier
Input	Reverse Bias	Reverse Bias	Cutoff	OFF Switch

Characteristics:

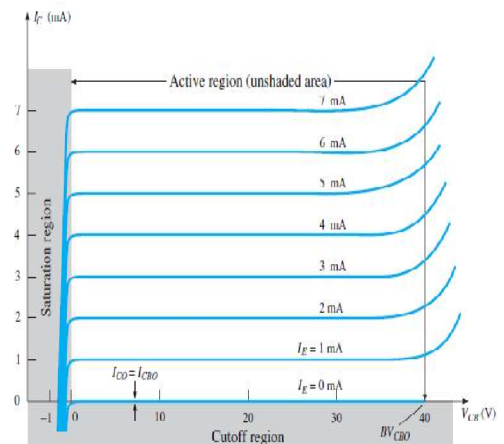
The plot of the input current against the input voltage with the output voltage as a parameter for a particular region of operation (Active).

- Graph is plotted between input voltage V_{BE} and current I_E keeping output voltage V_{CB} as a parameter.
- The current variation with the emitter to base voltage is similar to the forward characteristics of a p-n junction diode. However an increase in the magnitude of the collector to base voltage causes the emitter current to increase for a fixed V_{BE} because of the early effect or base width modulation.



Output Characteristics: The plot of the output current against the output voltage with the input current as a parameter.

- Graph is plotted between output voltage V_{CB} and output current I_C keeping input current I_E as a parameter.



- Active Region:

In this region J_{EB} is forward biased and J_{CB} is reverse biased. The collector current is independent of V_{CB} . It depends only on the emitter current I_E .

$$I_C = I_{\text{majority}} + I_{\text{minority}}$$

$$I_C = \alpha I_E + I_{CBO}$$

Where I_{CBO} (also noted as I_{CO}) is the leakage current called as collector to base leakage current when emitter is open. Since this current is very small in magnitude $I_C \approx I_E$.

- Cut off Region:

Here both the junctions are reverse biased. The region below $I_E=0$ characteristic is called as cut off region. In this region $I_C = I_{CBO}$ and in the range of nano amperes.

- Saturation Region:

Here both the junctions are forward biased. The region is to the left of the graph where V_{CB} is slightly positive and $I_E=0$. It gives an exponential variation in the collector current. J_{CB} is forward biased means collector is positive with respect to base. It gives rise to a hole current flowing from collector to base and is opposite to the original flow due to the transistor action. Output resistance of CB is very high because a very large change in collector voltage causes a very small change in collector current.

CB Current Amplification Factor:

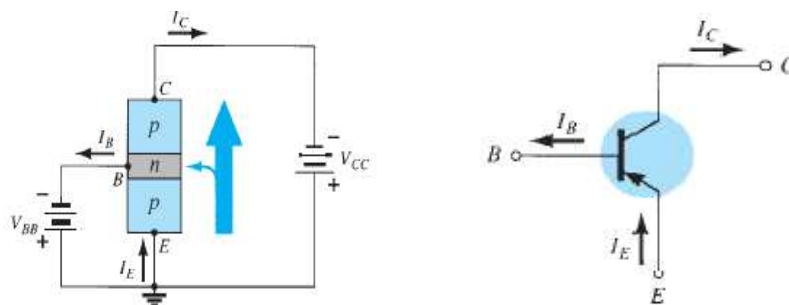
It is the ratio of output collector current to the input emitter current.

$$\text{DC current gain } \alpha_{dc} = \frac{I_C}{I_E} \quad \alpha \approx 1 \text{ (Practically 0.9 to 0.998)}$$

$$\text{AC current gain } \alpha_{ac} = \frac{\Delta I_C}{\Delta I_E} \text{ with } V_{CB} \text{ constant}$$

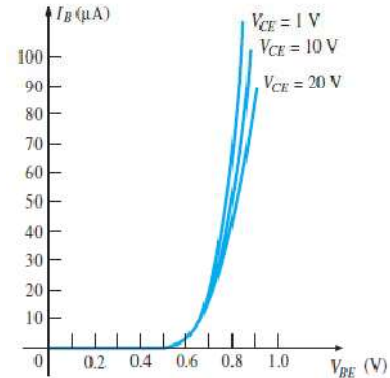
Common Emitter:

Here emitter terminal is common between the base and the collector.



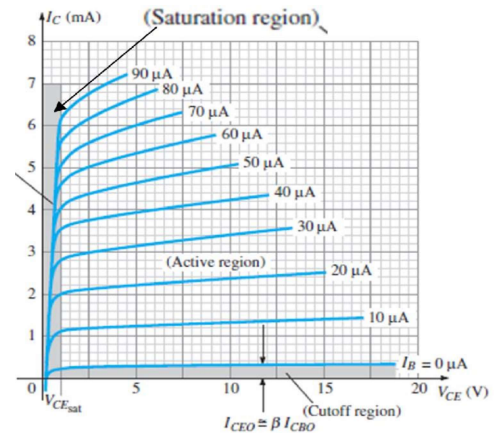
Input Characteristics:

- Graph is plotted between input voltage V_{BE} and input current I_B keeping output voltage V_{CE} as a parameter.
- Characteristics are similar to that of a forward biased diode. For a constant V_{BE} the magnitude of the base current decreases. With increasing V_{CE} . This is because increasing V_{CE} the effective base width and hence the recombination base current decreases.



Output Characteristics:

- Graph is plotted between output voltage V_{CE} and output current I_C keeping input current I_B as a parameter.
- Active Region:
In this region J_{EB} is forward biased and J_{CB} is reverse biased. Output characteristics in the active region are not horizontal lines because for a fixed value of I_B the magnitude of collector current increases with V_{CE} due to early effect.



$$I_C = \alpha I_E + I_{CBO} \quad (\text{CB})$$

$$= \alpha (I_E + I_E) + I_{CBO}$$

$$I_C = \alpha / (1 - \alpha) I_B + I_{CBO} / (1 - \alpha)$$

$$I_C = \beta I_B + I_{CEO}$$

Since the leakage current I_{CEO} is very small $I_C \approx \beta I_B$

- Cut off Region:
Here both the junctions are reverse biased. The region below $I_B = 0$ characteristic is called as cut off region. In this region $I_C = I_{CEO}$ (Collector to emitter leakage current with base open)
- Saturation Region:
Here both the junctions are forward biased by at least the cutin voltage. The current I_C is independent of I_B .

CE Current Amplification Factor:

It is the ratio of output collector current to the input baser current.

$$\text{DC current gain} \quad \beta_{dc} = \frac{I_C}{I_B} \quad \text{practically } 50-400$$

$$\text{AC current gain} \quad \beta_{ac} = \frac{\Delta I_C}{\Delta I_B} \quad \text{with } V_{CE} \text{ constant}$$

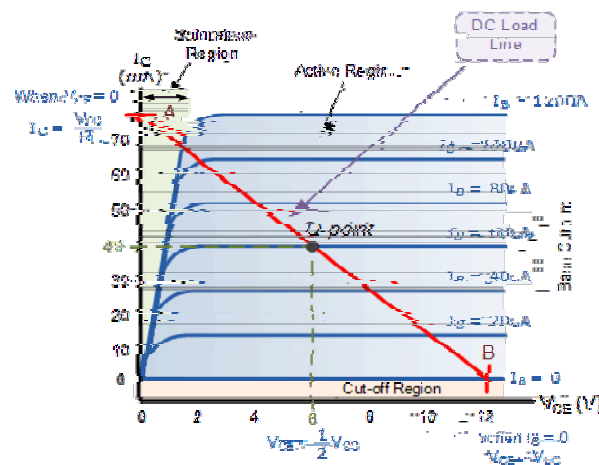
Relation between α and β :

$$I_E = I_C + I_B \quad \alpha = \frac{I_C}{I_E} \quad \text{and} \quad \beta = \frac{I_C}{I_B}$$

$$\frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta} \quad \alpha = \frac{\beta}{1 + \beta} \quad \beta = \frac{\alpha}{1 - \alpha}$$

DC Biasing:

Biasing is the application of external dc supply to establish a fixed level of current and voltage. Transistor operates only at a particular point of the characteristics called operating point or Q-point/ Quiescent point.

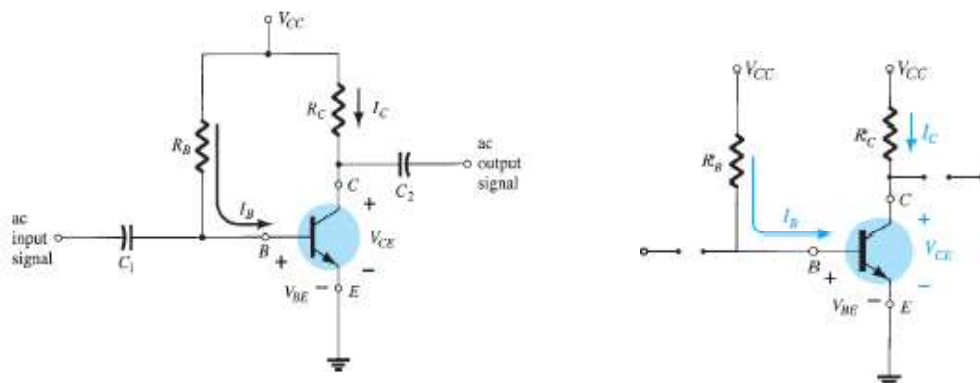


Types of Biasing

1. Fixed Bias Configuration
2. Fixed bias with emitter resistor
3. Voltage divider bias configuration
4. Collector feedback configuration

Fixed Bias Configuration:

It is the simplest transistor DC bias configuration using npn transistor. In the DC analysis capacitors are open circuited as shown in the figure.



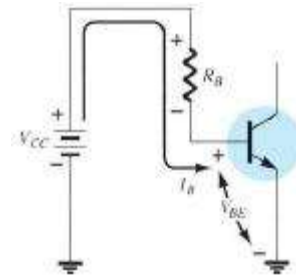
Base Emitter Loop:

Applying KVL we have

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

Since V_{BE} and V_{CC} are constant, the selection of a base resistor R_B sets the level of base current for the operating point.



Collector Emitter Loop:

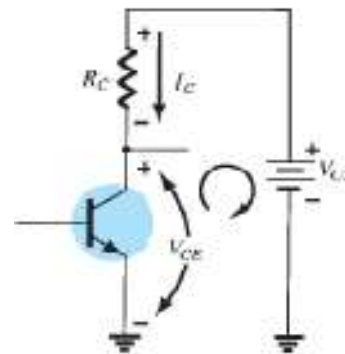
$$I_C = \beta I_B$$

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

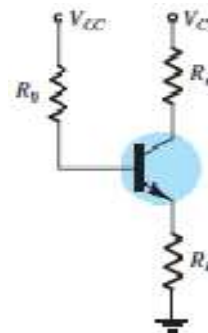
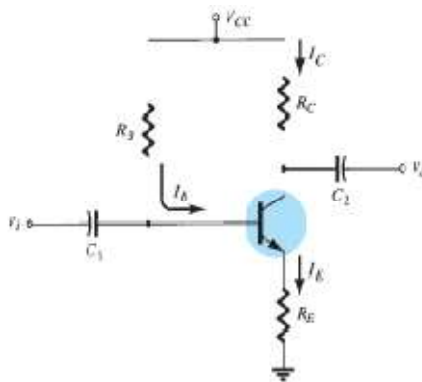
So R_B controls I_B and I_C , and the level of R_C determines the magnitude of V_{CE} .

$$V_{CE} = V_C - V_E \quad V_{BE} = V_B - V_E$$



Fixed Bias with emitter resistor Configuration:

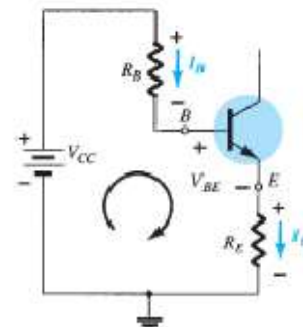
It contains an emitter resistor to improve the stability level over fixed bias.



Base Emitter Loop:

Applying KVL we have

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$



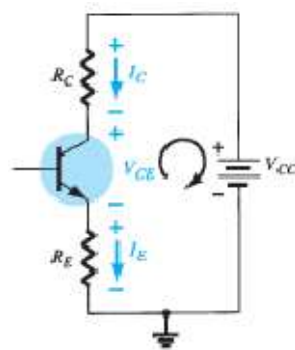
$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$

Collector Emitter Loop:

$$I_C = \beta I_B$$

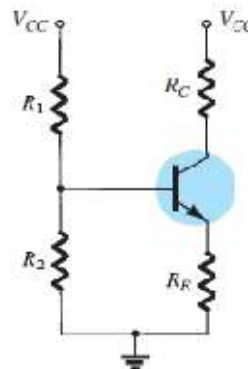
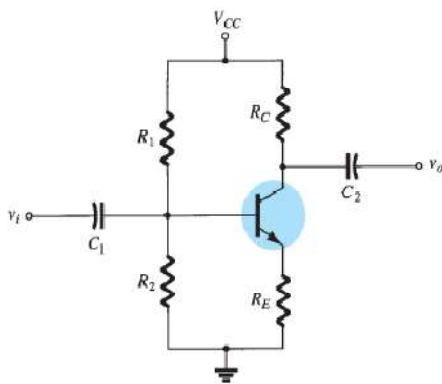
$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$



Voltage divider bias Configuration:

In the previous bias configurations the Q-points were dependent on β which is temperature sensitive. Voltage divider bias arrangement reduces the dependency of β .



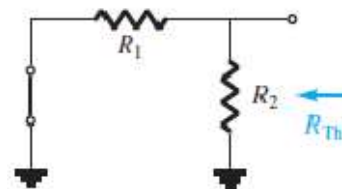
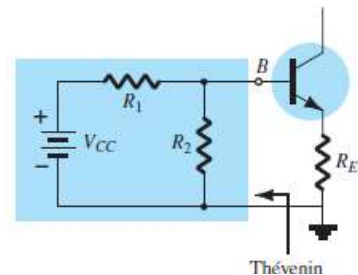
Exact Analysis:

The input side can be solved by Thevenin equivalent circuit.

For thevenin equivalent resistance the input dc source is short circuited.

$$R_{TH} = \frac{R_1 R_2}{R_1 + R_2}$$

For thevenin equivalent voltage the source V_{CC} is returned to the circuit and the voltage is given by



$$V_{TH} = V_{R2} = \frac{V_{CC}R_2}{R_1 + R_2}$$

The Thevenin equivalent circuit is redrawn and the IBQ can be found by applying KVL

$$V_{TH} - I_B R_{TH} - V_{BE} - I_E R_E = 0$$

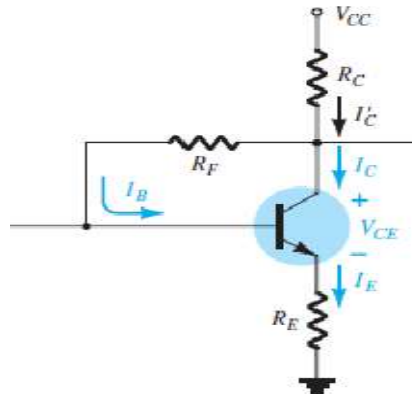
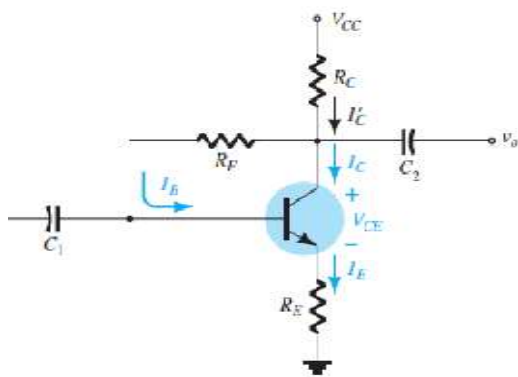
$$I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta + 1)R_E}$$

By solving the output collector emitter loop by KVL

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Collector feedback configuration:

An improved level of stability can be obtained by providing a feedback path.



Base Emitter Loop:

Applying KVL and solving we have

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)}$$

Collector Emitter Loop:

Applying KVL and solving we have

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

